

SEBASTIAN CRUZ

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EDUCATION AND CERTIFICATIONS

Middlebury College

Bachelor of Arts in Computer Science. College Scholar's List (Highest Academic Honors)

Middlebury, VT

Expected May 2027

QuestBridge Scholar. GPA: 3.82/4.00.

Relevant Coursework: Software Development, AI, Algorithms, Machine Learning

Certifications: CodePath Web Development (Intro & Intermediate), Google Project Management

EXPERIENCE

AI Lead and Backend Developer

Spring 2026 – Present

Altheros Capital

Remote

- Led backend AI team as technical lead, coordinating development efforts, reviewing pull requests, assigning tasks, and contributing to Agile sprint planning and architectural decisions.
- Designed and implemented Retrieval-Augmented Generation (RAG) infrastructure in Python, building document ingestion pipelines that process, chunk, embed, and store knowledge in PostgreSQL/pgvector to support AI-powered knowledge retrieval.
- Built RESTful backend APIs, data models, and AI services using Python and Pydantic, researching and evaluating new technologies to improve system scalability, maintainability, and clean architecture.

Backend Developer

Summer 2026

Text Maddie

Remote

- Developed backend workflows for AI-powered lead conversion agents, implementing Python logic that automated followups, appointment scheduling, CRM updates, and integrations across HubSpot, Slack, and messaging platforms.
- Engineered AI agent behaviors, workflows, and tool integrations to support natural customer conversations while triggering automated business processes and coordinating multi-agent interactions.
- Diagnosed and resolved production issues through log analysis, workflow tracing, and backend debugging to ensure reliable AI agent performance.

Machine Learning Research Assistant

Summer 2025

Middlebury College

Middlebury VT

- Implemented InfoNCE and RINCE loss functions for NPSV-deep, a machine learning project for structural variant genotyping using deep neural networks in PyTorch, improving embedding-based classification accuracy by 0.5-2.9% across 4 datasets and increasing non-reference concordance by up to 3%.
- Trained and evaluated deep learning models in PyTorch Lightning using TensorBoard, optimizing model parameters and maintaining working iterations via Git version control.

Data Science Research Assistant

Summer 2025

Town of Bristol

Bristol, VT

- Analyzed Vermont Department of Taxes data to identify Local Option Tax (LOT) adoption, category usage, and revenue trends across municipalities.
- Developed 2024 LOT revenue projections for Bristol by category, accounting for the state's 30% share and data gaps to identify potential underreported revenue.
- Compared LOT programs in Brandon, Middlebury, and Vergennes, synthesizing revenue allocation and local business feedback to assess economic impacts, transparency, and public support.

ADDITIONAL INFORMATION

Languages: Python, JavaScript, TypeScript, Java

Backend and AI/ML: FastAPI, Pydantic, PostgreSQL, RESTful APIs, RAG, LLM Agents, PyTorch

Frontend and Tools: React, Next.js, Git/GitHub, Agile/Scrum

Spoken Languages: English (Fluent), Spanish (Native), Portuguese (Intermediate)